Applied constraints in the two-singlet dark-matter scan

Collaborator discussion: implementation summary

29 October 2026

We list the constraints actually used in the $v_x = 0$ scan, the separate theoretical and thermal diagnostics, and the HiggsTools analyses selected in the reference campaign. The reference is `odysseus-100x1000-66662`, with HiggsTools 1.1.3, micrOMEGAs 7.1.4, BSMPT 3.2.1 and constraint profile `trsm_constraints_v2`. Its saved campaign and seed metadata were read directly from Odysseus. Both local Higgs datasets match the campaign's complete dataset checksums. The detailed derivations and source index remain in `docs/constraints-and-selection.tex`; this document provides the requested bullet list and analysis tables.

1 Model and meaning of an accepted point

- The SM-like scalar has $M_1 = 125.09\,\text{GeV}$. The dark scalar is $h_3 = X$, with $v_x = 0$ and zero doublet projection. With $\theta = a_{12}$, the visible couplings scale as $k_1 = \cos\theta$ and $k_2 = \sin\theta$; $a_{13} = a_{23} = 0$.
- In the current v2 selection, **full viability** means an explicit pass of tree theory, the experimental subset, flavour and the aggregate DM assessment:

$$V = T \wedge E \wedge F \wedge D.$$

  Here T is `thc`; E is `experimental_subset`; F is `flavour`; and D is `dm`. Non-DM viability is $T \wedge E \wedge F$. An unknown result cannot supply a required pass.
- The experimental subset combines HiggsBounds, HiggsSignals and the correlated S, T, U test, with the separate W-mass test added within its tabulated mass coverage. Flavour is evaluated separately.
- The original campaign stops each of 100 seeds after 1,000 points with both `evo=True` and `thc=True`. It stores all evaluated draws, including excluded and unassessed points. The resulting ledger contains 591,340 rows; the target of 100,000 counts tree-theory and RG-integration passes, rather than 100,000 fully viable points.
- The ordinary full-viability definition does not impose a global tree vacuum, RG boundedness, RG unitarity, a strong phase transition, or a LISA SNR threshold. Their separate flags must be requested explicitly when forming such subsets. The cumulative HB–HS–W–DM plotting sequence also omits tree theory, S, T, U and flavour, so its last stage is not V.

2 Theoretical constraints and separate theory assessments

- **Tree-level boundedness from below: required.** The potential convention is

$$V = \frac{1}{2}\sum_i \mu_i^2\varphi_i^2 + \frac{1}{4}\sum_i \lambda_i\varphi_i^4 + \frac{1}{4}\sum_{i<j} \lambda_{ij}\varphi_i^2\varphi_j^2, \qquad \varphi = (h, s, x).$$

Define $A_{ii} = \lambda_i$, $A_{ij} = \lambda_{ij}/2$ and $\bar{A}_{ij} = A_{ij} + \sqrt{A_{ii}A_{jj}}$. The implemented copositivity conditions are

$$A_{ii} \geq 0, \qquad \bar{A}_{12}, \bar{A}_{13}, \bar{A}_{23} \geq 0,$$
$$\sqrt{A_{11}A_{22}A_{33}} + A_{12}\sqrt{A_{33}} + A_{13}\sqrt{A_{22}} + A_{23}\sqrt{A_{11}} + \sqrt{2\bar{A}_{12}\bar{A}_{13}\bar{A}_{23}} \geq 0.$$

The numerical tolerance is 10^{-12} below zero; small negative diagonal values within that tolerance are set to zero in square roots.

- **Perturbative unitarity: required.** The strict bounds are $|\lambda_{\Phi,S,X}| < 4\pi$, $|\lambda_{\Phi S,\Phi X,SX}| < 8\pi$, and $|u_a| < 16\pi$ for every eigenvalue of

$$U = \begin{pmatrix} 12\lambda_\Phi & 2\lambda_{\Phi S} & 2\lambda_{\Phi X} \\ 2\lambda_{\Phi S} & 6\lambda_S & \lambda_{SX} \\ 2\lambda_{\Phi X} & \lambda_{SX} & 6\lambda_X \end{pmatrix}.$$

  Equality fails these strict bounds. The inputs must be finite and $v_s \neq 0$.

- **Unbounded tree potential: veto.** The independent vacuum assessment can identify an unbounded quartic direction or a flat quartic direction destabilised by its quadratic term. Either status also sets `thc=False`.

- **Global tree vacuum: separate assessment.** The desired vacuum $(v, v_s, 0)$ is compared with minima on all eight field supports, including the origin. It must be local and satisfy $V_{\text{target}} - V_{\text{min}} \leq 10^{-8}\max(1, |V_{\text{target}}|, |V_{\text{min}}|)$, using numerical potentials in GeV^4; degenerate minima pass. Singular stationary systems and flat directions receive dedicated checks. An unresolved case is unknown. A deeper bounded minimum does not by itself fail ordinary full viability.

- **One-loop RG evolution to 1 TeV: separate assessments.** Running begins at 91 GeV, with $(g_3, g_2, g_1, y_t) = (1.221, \sqrt{0.424}, \sqrt{0.1273}, 0.96738)$. The couplings are screened for BFB and the same unitarity conditions on solver nodes and 513 logarithmically spaced scales. The integration uses relative/absolute tolerances $10^{-8}/10^{-10}$; violation brackets are refined to 10^{-5} GeV. No threshold or precision pole-to-running matching is added. `evo` records integration success, not physical acceptance of the entire running trajectory.

- **Strict theory subset: optional.** This combines the global tree-vacuum verdict with RG integration success, RG BFB and RG unitarity. It is available for comparisons but is not a factor in V.

- **Positive quartic determinant: optional EWPT preselection.** The `--ewpt-require-eq418` option requires positive diagonal entries, positive two-by-two principal minors and $\det A > 0$. It was *disabled* in the reference run and remains disabled in the two new profiles. The current determinant uses $A_{ii} = \lambda_i$, $A_{ij} = \lambda_{ij}/2$; this is stronger than copositivity. The plot itself applies no determinant cut.

Implementation: `test_trsm_theory_constraints.py`, `trsm_theory_diagnostics.py`, `trsm_inputs.py`, `generate_trsm_points.py` and `test_trsm_ewpt.py`.

3 Experimental constraints

- **HiggsBounds: required.** HiggsTools evaluates applicable search limits with the point's masses, production inputs, widths and branching fractions. For each scalar it selects the most sensitive analysis using expected sensitivity, then tests its observed limit. The pipeline uses the provider's

`allowed` verdict, rather than intersecting every displayed observed ratio. See the HiggsBounds API. Section 7 lists the 31 different IDs selected somewhere in the reference ledger; rows with a zero selection count are omitted.

- **HiggsSignals: required.** The implemented condition is $\chi^2_{\text{point}} - \chi^2_{\text{SM}} < 4.00$. The SM reference uses the same M_1 and SM decay-input pipeline. All 22 installed measurement files, containing 159 submeasurements, enter the calculation. This fixed threshold is a screening prescription, not a profiled confidence region for every scan parameter. A nonfinite likelihood is unassessed.

- **Visible, invisible and scalar-cascade decays: included in HiggsTools.** The independent invisible widths are

$$\Gamma(h_i \to h_3 h_3) = \frac{K_{i33}^2}{8\pi M_i}\sqrt{1 - \frac{4M_3^2}{M_i^2}}, \qquad M_i > 2M_3,$$

  and zero at or below threshold. Here K_{i33} is the coefficient of $h_i h_3^2$ in the potential. Physical total widths include these channels and open $h_1 \to h_2 h_2$ or $h_2 \to h_1 h_1$ decays; the visible branching fractions are diluted consistently. There is no additional universal cut on the invisible Higgs branching fraction. A $10^{-10}\,\text{GeV}$ floor is used only in nonzero HiggsTools width records to avoid numerical zeroing.

- **Scope of the HiggsTools production input.** The adapter uses `scaledSMlikeEffCouplings` for h_1, h_2 and supplies the physical decay inputs above. Provider nonresonant identical-scalar pair-production estimates are therefore included where applicable; the adapter does not supply a model-specific self-coupling override to those estimates. Separately computed MadGraph hh and associated-production rates are not fed back into the stored HiggsBounds decision. Scalar-to-Z-scalar, flavour-changing top and charged-scalar channels do not acquire nonzero model inputs here.

- **Correlated electroweak precision observables: required.** With U allowed to vary, the scalar contributions obey $\Delta O = \sin^2\theta[O(M_2) - O(M_1)]$ for $O = S, T, U$. The cut is

$$(\boldsymbol{\Delta} - \boldsymbol{c})^T \Sigma^{-1} (\boldsymbol{\Delta} - \boldsymbol{c}) \leq 7.82, \qquad \Sigma_{ij} = \rho_{ij}\sigma_i\sigma_j,$$

  with

$$\boldsymbol{c} = (-0.045235, 0.013373, -0.016131),$$

$$\boldsymbol{\sigma} = (0.094703, 0.11518, 0.087103), \qquad \rho = \begin{pmatrix} 1 & 0.048 & 0.747 \\ 0.048 & 1 & -0.162 \\ 0.747 & -0.162 & 1 \end{pmatrix}.$$

  The repository attributes these fit inputs to Jens Erler, private communication, 16 May 2025. They describe the implemented fit.

- W **mass: required within coverage.** For $133 \leq M_2/\text{GeV} \leq 999$, require $|\sin\theta| \leq s^W_{\max}(M_2)$ using the cubic interpolation of `datafiles/Tania_MW_SnowmassWhitepaper.dat`. Equality passes. There is no extrapolation; outside coverage this factor is a non-veto and is omitted from the v2 experimental subset. The S, T, U and W tests are separate screens, without a joint likelihood.

- **Flavour: required separately.** For each visible scalar and $\ell = \mu, \tau$, compare $\text{BR}(\Upsilon(1S) \to \gamma h_i)\,\text{BR}(h_i \to \ell^+\ell^-)$ with the strongest available individual Belle/BaBar 90% CL curve. The nominal windows are $[0.212, 9.2]\,\text{GeV}$ and $[3.55, 9.2]\,\text{GeV}$, respectively, also restricted to each digitised curve's coverage. The supplied tau curves start near $3.83\,\text{GeV}$. Interpolation is linear in mass and logarithmic in the limit, with no extrapolation. Covered ratios must be at most one. These prompt, narrow-resonance limits use the implemented production factor $F_{bS} = \frac{2}{3}[1 - (M_i/m_\Upsilon)^6]$,

$m_b = 4.18\,\mathrm{GeV}$, $m_\Upsilon = 9.46040\,\mathrm{GeV}$, $\alpha^{-1} = 137.035999084$ and $\mathrm{BR}(\Upsilon \to e^+e^-) = 0.0239$. There is no extra lifetime acceptance cut. Outside coverage or at zero production, the test is a non-veto; missing required decay information is unassessed. In this branch it constrains a light h_2. The source identified by the implementation is Belle, arXiv:2112.11852.

Implementation: `test_trsm_higgstools.py`, `generate_trsm_info.py`, `singlet_EWPO.py`, `trsm_constraint_profile.py` and `trsm_flavour.py`.

4 Dark-matter constraints

- **Successful micrOMEGAs calculation: required.** The reference backend is 7.1.4. Solver failures or invalid physical outputs give an unknown aggregate DM verdict. Virtual W/Z decays are enabled in the reference configuration. The abundance fraction is $\xi = \min(1, \Omega h^2/0.12)$; scattering predictions scale with ξ and annihilation signals with ξ^2.

- **Relic abundance: required upper bound.** Require $0 \le \Omega h^2 \le 0.121$. Underabundance is allowed; there is no positive lower bound and no requirement that h_3 supply all DM. The 0.12 rescaling reference differs from the 0.121 acceptance limit. The freeze-out diagnostic is $T_f = M_3/X_f$ when $X_f > 0$ is available.

- **Spin-independent direct detection: required.** The default is the LZ WS2024 observed power-constrained 90% CL elastic isoscalar per-nucleon limit, from the published numerical table associated with arXiv:2410.17036. The code compares the micrOMEGAs neutron SI cross section with
$$\xi\,\sigma_n^{\mathrm{SI}} \le L_{\mathrm{LZ}}(M_3).$$
The 26-point table covers 9–10000 GeV, with log–log interpolation and $1\,\mathrm{cm}^2 = 10^{36}\,\mathrm{pb}$. No extrapolation is performed. For positive ξ, points below 9 GeV have an unavailable DD assessment and cannot supply a DM pass. At $\xi = 0$, finite scattering is a non-veto. `dm_dir_det` is the unscaled cross section and `dm_dir_det_limit` is L_{LZ}/ξ; the comparison must not rescale both again. The optional LZ2026 approximation is not used in this campaign.

- **Gamma-ray lines: exclusion screen.** Compare each usable line flux with the stored Fermi-LAT R16 limit:
$$\max_a \frac{\xi^2 \Phi_a}{L_{\mathrm{R16}}(E_{\gamma,a})} \le 1.$$
The tabulated energy range is 0.214–462 GeV, with log–log interpolation and no extrapolation. The line energies are $E_\gamma = M_3$ for $\gamma\gamma$ and $M_3(1 - m_Y^2/(4M_3^2))$ for γY. The driver integrates 0.5° patches inside a 16° circular region, excluding $|b| < 5^\circ$ when $|l| > 6^\circ$, and includes photon multiplicity. It uses the largest individual ratio rather than combining nearby lines. This is the implemented line-flux approximation, rather than a full ROI likelihood. With no usable line, the result is a documented non-veto. The printed continuum flux supplies no additional exclusion.

- **CMB energy injection: enabled and required.** The reference uses the built-in Planck2018 spectral-deposition calculation with $p_{\mathrm{ann}}^{\mathrm{max}} = 3.2 \times 10^{-28}\,\mathrm{cm^3 s^{-1} GeV^{-1}}$ and $\xi^2 R_{\mathrm{CMB}}^{\mathrm{raw}} \le 1$. This is the micrOMEGAs low-velocity s-wave approximation; the spectrum uses the default $v_0 = \sqrt{3}\,v_{\mathrm{rot}}/c$, with $v_{\mathrm{rot}} = 220\,\mathrm{km/s}$. An unavailable enabled CMB calculation cannot supply a pass. The Planck 2018 cosmological-parameter analysis provides the observational context.

- **Aggregate DM decision.** With a successful solver, combine the relic, direct-detection and enabled CMB passes with the absence of a line exclusion. A known exclusion makes `dm=False`; otherwise a missing required result gives `dm=None`. Equality at the listed upper limits passes. The plotting suite uses the stored aggregate decision.

Implementation: `test_trsm_DM.py`, `trsm_direct_detection.py`, `trsm_cmb.py`, `DM/main.c` and `DM/data/lz2024/observed-si-v2.json`.

5 Thermal diagnostics and interpretation of plot 97

- **EWPT entry gate.** Run BSMPT when tree theory passes and the experimental subset is not explicitly false. The general gate permits an unknown experimental assessment, and does not require DM or flavour. The reference campaign has 31,204 eligible points; all explicitly pass the experimental and flavour tests.

- **Stars in plot 34b and CalcGW plots.** A star denotes a successful stored EW-entry diagnostic with $|w_{1,\mathrm{false}}| < 5\,\mathrm{GeV} \leq |w_{1,\mathrm{true}}|$ and $\Delta v_{\mathrm{EW}}(T_c)/T_c > 1$. Equality at one does not qualify. Every transition marker from that scan point has a star in the CalcGW panels. A star identifies the parent point; the particular transition producing the plotted GW signal need not itself be EW entry.

- **Qualitative thermal flags: separate from viability.** The baryogenesis-candidate flag requires the strong critical EW entry above. The qualitative GW-candidate flag requires an available FOPT with total field-space jump $\Delta\phi/T > 1$ at T_c, T_n or T_p. These flags do not calculate a baryon asymmetry or determine a LISA signal. Percolation, completion, equilibrium Z_2 restoration near freeze-out, VEV changes over $[T_f/2, 2T_f]$, and corresponding coupling proxies are stored diagnostics; they are not additional cuts in V.

- **Native CalcGW predictions.** The saved enrichment evaluates the EWPT-eligible points with $T_{\max} = 1000\,\mathrm{GeV}$, $T_* = T_p$, $v_w = 0.95$ and $\epsilon_{\mathrm{turb}} = 0.1$. A plotted pair must have successful native GW and requested-temperature statuses and successful NLO stability, tracing and coexistence assessments, with finite $\alpha > 0$, $T_* > 0$ and $\beta/H_* \geq 1$. Nonpositive values cannot appear on logarithmic axes. Failures are not zero-signal predictions. All successful phase pairs are shown, without selecting a cosmological transition history. See the BSMPT v3 paper.

- **Plot 97 includes experimental constraints in this campaign.** There are 20,763 points with 34,227 successful transition pairs. Every such point passes `thc`, HiggsBounds, HiggsSignals, EWPO, the applicable W test and flavour. The all-points CalcGW panel adds no further full-viability cut. The version labelled full viability leaves 431 points and 647 transition pairs. Among the other successful points, 19,927 fail DM and 405 have unknown DM status.

- **LISA SNR interpretation.** The stored total-spectrum SNR uses three years of integration and a sensitivity-weighted frequency integral of the squared GW spectrum. The line SNR$= 5$ is the requested sensitivity reference; it is neither an additional scan cut nor a claim of a five-standard-deviation discovery. The new plot 97b shows the same values with $1 \leq \mathrm{SNR} \leq 100$, in both selections.

- **Result in the saved sample.** No transition reaches SNR$= 5$. The largest SNR is 1.5344, at $M_2 = 401.3081\,\mathrm{GeV}$ and $M_3 = 62.5575\,\mathrm{GeV}$; this point has a star and fails DM. It is the only transition in the 1–100 view. The largest fully viable SNR is 4.4912×10^{-11}, so that zoom is empty. Seven successful-GW points carry the plot-34b star; none pass full viability.

Implementation and evidence: `trsm_gw.py`, `reprocess_trsm_gw.py`, `plot_trsm_constraint_suite.py` and the source audit, reference metadata and LISA summary in `validation/collaborator-update-20261029/`.

6 The two new scan profiles

- Profile A: independent $M_2, M_3 \in [4, 1000]\,\mathrm{GeV}$; `config/odysseus-m2-m3-4-1000.json`.
- Profile B: independent $M_2 \in [4, 200]\,\mathrm{GeV}$ and $M_3 \in [4, 65]\,\mathrm{GeV}$; `config/odysseus-m2-4-200-m3-4-65.json`.
- Both use the reference seeds 66662–66761 in new output directories, 100 concurrent workers, and 1,000 `evo&thc` passes per seed. They preserve $v_s \in [50, 500]\,\mathrm{GeV}$, $k_1 \in [0.9, 1]$ with both mixing-angle signs, $\lambda_X \in [0, 10]$, and independent signs with uniform $\log_{10}(|K_{133}|/\mathrm{GeV}) \in [-4, 3]$ and $\log_{10}(|K_{233}|/\mathrm{GeV}) \in [-3, 5]$. The portal quartics are derived from these trilinears.
- EWPT and the original MadGraph channels `gg_heta0` and `pp_eta0Z` remain enabled, with MadGraph restricted to full viability. Workers use one native thread each. CalcGW and the additional hh and loop-induced Zh_2 enrichments remain separate saved-point operations, as in the original campaign. The launch commands are in `docs/parallel-campaigns.md`.

7 HiggsTools analyses in the installed datasets

The installed datasets contain 310 HiggsBounds limits and 22 HiggsSignals measurement files. The HiggsBounds table lists the 31 IDs with a nonzero selection count in the reference campaign; rows with a zero count are omitted. All HiggsSignals measurement files are listed. Each row is one provider ID; several rows may refer to the same experimental paper. These are the datasets used by the reference campaign, rather than a claim to include later experimental results.

- `hbdataset` revision `3539f9adf3724f8752701208bf059d1658824cd2`.
  SHA256: `c7e4070faac4e5846cb89a7f3cc7718686e58b82604f0c6659b93a0d2a39d0b2`.
- `hsdataset` revision `3c6c2538d9ba0549ea0559cf9b38c18eb80c3af1`.
  SHA256: `36076ada7b6dbd807b34e813c047d6f0859cfbcf5bcef91f752bcc9ba95496f3`.
- Scope A: the input topology can be nonzero in this adapter. Mass, width, kinematic and model-likeness conditions still determine applicability at each point; an A label does not assert sensitivity or exclusion throughout either scan rectangle.
- The domain is HiggsTools' own grid description. Masses are in GeV; width coordinates and other qualifications follow the linked implementation. Long decimal endpoints are displayed to six significant figures; the exported JSON catalogue keeps the original strings. The table does not extrapolate a limit outside its applicability region.
- Process notation follows HiggsTools: H denotes a neutral scalar; ggH and vbfH denote gluon fusion and vector-boson fusion; HW, HZ, Htt and bbH denote associated production; eeHZ denotes LEP Higgsstrahlung; inv denotes invisible decay; gam denotes a photon. The symbol $>$ separates production and decay, and comma-separated entries form an analysis combination.
- X1, X2 and X3 and the corresponding M1, M2 and M3 in the provider descriptions label the parent and daughter roles of that search. They are not fixed identifications with the model's h_1, h_2, h_3 or their masses. HiggsBounds tests allowed particle assignments.

The final column counts distinct reference-campaign points for which the analysis was selected for at least one scalar by expected sensitivity, using all 591,340 saved rows. A zero count does not mean a search is inapplicable; only selected limits, not every tested limit, are recorded exhaustively. There are 31 different selected IDs.

7.1 HiggsBounds searches selected in the reference campaign

ID / scope	Search process	Provider domain	Experiment / $\mathcal{L}$ [fb^{-1}]	Reference	Selected
220910910 A	LHC13[H]->X1->(X2->[tautau])(X3->[bb])	M1=(251,1600) M2=M3=(125,125)	ATLAS 139	2209.10910	20,420
231115956 A	LHC13[H]->X1->(X2->[bb])(X3->[bb,tautau,gamgam])	M1=(251,5000) M2=M3=(125,125)	ATLAS 139	2311.15956	21,248
240213358 A	LHC13[H]->X1->(X2->[bb])(X3->[mumu,tautau])	M1=(125,125) M2=M3=(12,60)	CMS 138	2402.13358	18,813
240316926 A	LHC13[H]->X1->(X2->[bb])(X3->[bb,tautau,gamgam,WW])	M1=(260,4000) M2=M3=(125,125)	CMS 138	2403.16926	48,638
2110003131 A	LHC13[H,vbfH,HZ,HW]->X1->(X2->[mumu])(X3->[bb])	M1=(125,125) M2=M3=(16,62)	ATLAS 139	2110.00313	3
151006534 A	LHC8[ggH]->X1->(X2->[tautau])(X3->[tautau])	M1=(125,125) M2=M3=(4,8)	CMS 19.7	1510.06534	1,114
13003 A	LHC8[H>WW]	M=(110,600)	CMS 24.4	CMS-PAS-HIG-13-003	1,333
13121129 A	LHC8[ggH,vbfH,HW,HZ]>[WW]	M=(110,600)	CMS 24.3	1312.1129	4,897
2109060552 A	LHC13[vbfH>WW]	M=(1000,4500)	CMS 137	2109.06055	8,664
1804019391 A	LHC13[vbfH,H]>[ZZ]	M=(130,3000) Gam/M=(0,0.3)	CMS 35.9	1804.01939	11,548
13125353 A	LHC8[vbfH,HW,Htt,H,HZ]>[ZZ]	M=(110,1000)	CMS 24.8	1312.5353	268
2009147911 A	LHC13[ggH>ZZ]	M=(400,2000) Gam/M=(0,0.15)	ATLAS 139	2009.14791	40,736
2009147912 A	LHC13[ggH>ZZ]	M=(210,400)	ATLAS 139	2009.14791	73,798
206022 A	LEP[eeHZ>none]	M=(1,100)	OPAL 0.77	hep-ex/0206022	3,294
14096212 A	LHC8[HW,HZ]>[bb]	M=(110,140)	ATLAS 25	1409.6212	5,271
3060332 A	LEP[eeHZ>bb]	M=(12,120)	LEPComb 2.5	hep-ex/0306033	14,363
1810118221 A	LHC13[ggH>bb]	M=(50,350)	CMS 35.9	1810.11822	35

ID / scope	Search process	Provider domain	Experiment / $\mathcal{L}$ [fb^{-1}]	Reference	Selected
1808023801 A	LHC13[H]>[WW,ZZ]	M=(300,3000)	ATLAS 36.1	1808.02380	103,650
150400936 A	LHC8[vbfH,HW,Htt,H,HZ]>[WW,ZZ]	M=(145,1000)	CMS 24.8	1504.00936	2,398
2004146362 A	LHC13[vbfH]>[WW,ZZ]	M=(300,5000)	ATLAS 139	2004.14636	46,146
12045 A	LHC8[vbfH,HW,Htt,H,HZ]>[bb,tautau,WW,ZZ,gamgam]	M=(110,600)	CMS 17.3	CMS-PAS-HIG-12-045	337,338
240518149 A	LHC13[vbfH,HW,Htt,H,HZ]>[gamgam]	M=(70,110)	CMS 132	2405.18149	12,827
230110731 A	LHC13[HW,Htt,H,HZ]>[inv]	M=(125,125)	ATLAS 164	2301.10731	215,728
1809059372 A	LHC13[vbfH,H]>[inv]	M=(110,1000)	CMS 35.9	1809.05937	132,490
1809059371 A	LHC13[vbfH,HW,H,HZ]>[inv]	M=(125,125)	CMS 62.8	1809.05937	7,021
501033 A	LEP[eeHZ>inv]	M=(50,110)	L3 0.63	hep-ex/0501033	4,996
107032 A	LEP[eeHZ>inv]	M=(90,118)	LEPComb 0.87	hep-ex/0107032	2,206
7070373 A	LEP[eeHZ>inv]	M=(5,115)	OPAL 0.66	0707.0373	4,969
180906682 A	LHC13[vbfH>inv]	M=(75,3000)	ATLAS 36.1	1809.06682	145
180306553 A	2dlikelihood{LHC13[ggH>tautau],LHC13[bbH>tautau]}	M=(90,3200)	CMS 35.9	1803.06553	85
14029 A	2dlikelihood{LHC8[ggH>tautau],LHC8[bbH>tautau]}	M=(90,1000)	CMS 24.6	CMS-PAS-HIG-14-029	1,297

7.2 HiggsSignals measurements

All files below enter the HiggsSignals likelihood. The submeasurement count identifies the number of channels or bins in each file; correlations are supplied by the dataset.

ID	Measurement file / channel	Experiment / $\mathcal{L}$ [fb^{-1}]	Reference	Bins
180809054	h125/WW_H_LHC13_ATLAS_36.json	ATLAS LHC13 36.1	1808.09054	2
200701984	h125/WW_H_LHC13_CMS_137.json	CMS LHC13 137	2007.01984	5
190310052	h125/WW_VH_LHC13_ATLAS_36.json	ATLAS LHC13 36.1	1903.10052	2
200403447	h125/ZZ_4l_STXS_LHC13_ATLAS_139.json	ATLAS LHC13 139	2004.03447	12
210304956	h125/ZZ_4l_STXS_LHC13_CMS_137.json	CMS LHC13 137	2103.04956	19
211106712	h125/bb_Htt_LHC13_ATLAS_139.json	ATLAS LHC13 139	2111.06712	1
180808242	h125/bb_LHC13_CMS_102.json	CMS LHC13 102.1	1808.08242	5
200702873	h125/bb_VH_LHC13_ATLAS_139.json	ATLAS LHC13 139	2007.02873	5
201108280	h125/bb_vbfH_LHC13_ATLAS_126.json	ATLAS LHC13 126	2011.08280	1
2021021	h125/cc_LHC13_ATLAS_139.json	ATLAS LHC13 139	ATLAS-CONF-2021-021	1
191201662	h125/cc_LHC13_CMS_36.json	CMS LHC13 35.9	1912.01662	2
220700348	h125/gamgam_STXS_LHC13_ATLAS_139.json	ATLAS LHC13 139	2207.00348	28
210306956	h125/gamgam_STXS_LHC13_CMS_137.json	CMS LHC13 137	2103.06956	17
1812251	h125/mass_LHC13_LHCComb_36.json	LHCComb LHC13 36	datasetentry	1
200707830	h125/mumu_LHC13_ATLAS_139.json	ATLAS LHC13 139	2007.07830	1

ID	Measurement file / channel	Experiment / $\mathcal{L}$ [fb^{-1}]	Reference	Bins
200904363	h125/mumu_LHC13_CMS_137.json	CMS LHC13 137	2009.04363	4
160602266	h125/run1_rates_LHC8_LHCComb_25.json	LHCComb LHC8 25	1606.02266	20
220412957	h125/tautau_CMS_STXS_139ifb_LHC13_CMS_138.json	CMS LHC13 138	2204.12957	16
211004836	h125/tautau_CP_LHC13_CMS_137.json	CMS LHC13 137	2110.04836	3
181108856	h125/tautau_H_LHC13_ATLAS_36.json	ATLAS LHC13 36.1	1811.08856	2
2019045	h125/ttH_lep_LHC13_ATLAS_80.json	ATLAS LHC13 80	ATLAS-CONF-2019-045	6
180305485	h125/ttH_lep_LHC13_CMS_36.json	CMS LHC13 35.9	1803.05485	6